

CELLULAR LOCALIZATION OF INSULIN-LIKE IMMUNOREACTIVITY IN THE CENTRAL NERVOUS SYSTEM OF SPINY MICE, C57B16J AND C57B1KSJ MICE*

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The regional and cellular distribution of insulin-like immunoreactivity throughout the CNS of three rodent species were investigated using an immunofluorescence technique. No principal differences in the content and distribution of insulin-like material were revealed between lean animals and mice suffering from a genetically determined diabetes mellitus.

Since 1978 evidence has been accumulated suggesting that not only insulin-specific receptors are present in the brain but also the hormone itself (1, 3, 4). Although the average concentration of insulin in nervous tissue appears to be fairly high (25 fold higher than in the blood), there are great regional differences (3). The aim of the present investigation was to get more detailed information on the distribution of insulin-containing elements in the CNS of three rodent strains and to find out whether there is an effect of manifest diabetes mellitus on the content and/or localization of the hormone in the brain.

MATERIAL AND METHOD

Ten spiny mice (*Acomys cahirinus*), five lean mice of the strain C57B16J, five lean mice of the strain C57B1KSJ and five diabetic animals of the strain C57B1KSJ (db/db) were used throughout this study. Prior to decapitation, blood glucose levels and blood immunoreactive insulin concentrations were estimated as described earlier (2). Immunocytochemical procedures were carried out as follows: Deparaffinized 6 μ m thick coronal brain sections were treated with antiserum to insulin generated in guinea pigs (diluted 1 : 2 with phosphate buffered saline) for 1 hr. After repeated washing in PBS, the tissue was exposed to anti-guinea pig IgG prepared in goat, conjugated to FITC, for 2 hr. Then the slices were washed again and counterstained with Evans blue dye (1% for 2–3 min).

Specificity controls performed were as follows: Treatment of the tissue with antiserum previously adsorbed with an excess of antigen, use of highly diluted

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antisera, omission of the specific antiserum. All these procedures resulted in the disappearance of fluorescence. As a positive reference, the localization of a strong immunofluorescence to B cells of the islets of Langerhans was regarded. The insulin-induced immunofluorescence was analysed under a fluorescence microscope as a blind study, i.e., it was unknown to the investigator whether an animal was diabetic.

RESULTS

In general, nearly all brain regions studied showed a more or less pronounced immunofluorescence. High amounts of the hormone were present in some nuclei of the hypothalamus (Nuc. ventromedialis pars dorsalis, Nuc. infundibularis, Nuc. paraventricularis, Nuc. periventricularis, Nuc. arcuatus, Nuc. anterior hypothalami), the Corpus amygdaloideum, Nuc. lentiformis, different areas of the thalamus, frontal cortex (especially lamina VI), parietal cortex, allocortex, pyramidal cells of the hippocampus. Often, strongly reacting cells occurred in the immediate neighborhood of non-reactive neurons (indicated by the counterstaining procedure, see microphotos). Among the animals investigated were individuals nearly lacking in demonstrable immunoreactive material. It must be emphasized that these animals were not identical with the group of identified diabetic ones. On the other hand, among the diabetic mice were cases with a relatively high insulin level. In the nerve cells, the fluorescence was almost restricted to the nerve cell cytoplasm. Insulin-containing fibres were rather seldom (near the III. ventricle). Within the neurons, the reaction product formed characteristic circular structures surrounding the cell nucleus. In a few blood vessels, a weak reaction took place.

DISCUSSION

The distribution of insulin-like material as observed here does not differ from that demonstrated in other rodents by immunocytochemical (1) and radioimmunological (3) methods. Our earlier conjecture that the immunofluorescence might also be due to a component in the brain, differing in its structure and properties from native insulin, was corrected by data from the literature, convincingly indicating that insulin or a substance, extremely similar to the hormone, occurs in the CNS (3, 4). The physiological function of the hormone in the nervous tissue remains in speculation.

Van Houten *et al.* (5) have revealed some parallels between the intracerebral distribution of insulin-binding sites and the status of glucose metabolism.

FIG. 1. A strong reaction was observed in the hypothalamus (Nuc. ventromedialis and Nuc. infundibularis). A faint staining appears in the ependymal cells of the III. ventricle. $\times 120$

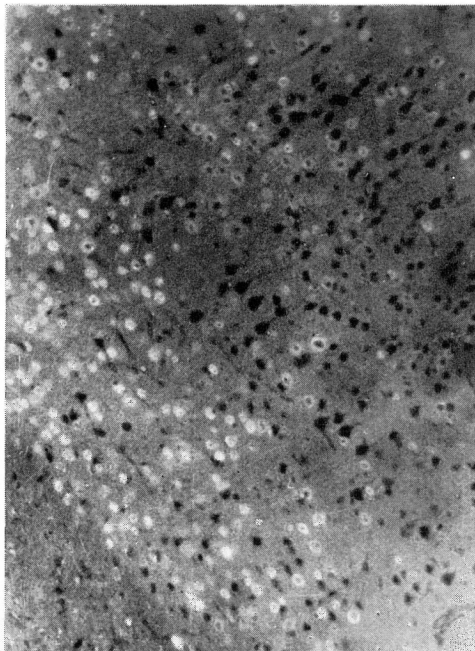
FIG. 2. Insulin immunoreactivity in the frontal cortex. Note non-reactive pyramidal neurons of the lamina III and V. $\times 120$

FIG. 3. Immunofluorescence to insulin-antibodies in pyramidal cells of the hippocampus CA₂ region. $\times 250$

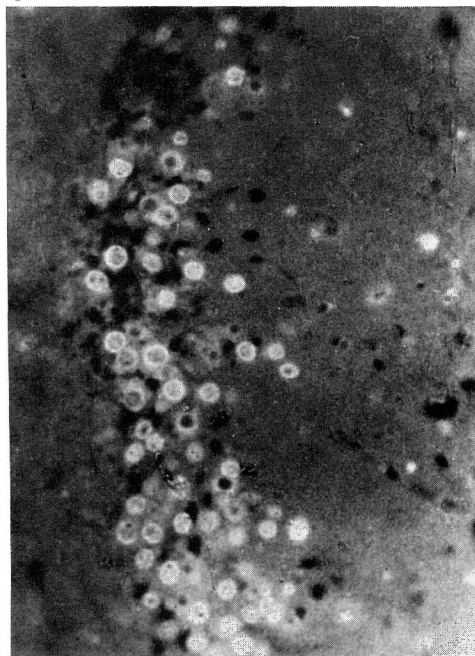
FIG. 4. A distinct nerve cell population in the area of Nuc. lentiformis displaying a strong reaction. $\times 400$



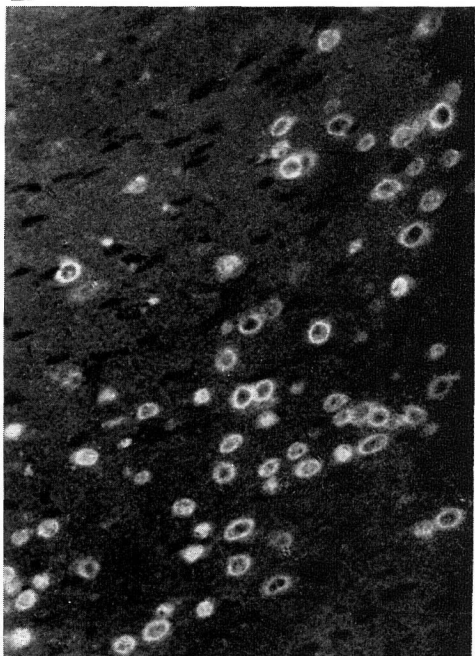
1



2



3



4

Havrankova and colleagues (3, 4) suppose a role of brain-located insulin independent of its function in carbohydrate regulation. In support of this, they published data on diabetic animals which did not suffer from a decreased insulin content in the CNS. Our own observations seem to confirm this, but they should be regarded as preliminary, since these findings are rather of qualitative than of quantitative nature. We are now about to study the question of a possible correlation between diabetes and insulin in the CNS by a micro-immunoassay.

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REFERENCES

1. Dorn, A., Bernstein, H.-G., Hahn, H.-J., Kostmann, G. and Ziegler, M.: An immunofluorescent reaction appears against insulin-antibodies in different CNS regions of two rat species. *Acta histochem.* (Jena) in press.
2. Dorn, A., Bernstein, H.-G., Hahn, H.-J. and Kostmann, G.: Are certain peculiarities in the distribution of somatostatin-like immunoreactivity (SLI) in the brain of sand rats (*Psammomys obesus*) possibly connected with an altered carbohydrate metabolism? *Acta histochem.* (Jena) 66; 168-173, 1980.
3. Havrankova, J., Schmechel, D., Roth, J. and Brownstein, M.: Identification of insulin in the rat brain. *Proc. Natl. Acad. Sci. (USA)* 75; 5737-5741, 1978.
4. Havrankova, J., Roth, J. and Brownstein, M.: Concentrations of insulin and insulin receptors in the brain are independent of peripheral insulin. Studies of obese and streptozotocin-treated rodents. *J. Clin. Invest.* 64; 636-642, 1979.
5. Van Houten, M., Posner, B. I., Kopriwa, B. W. and Brawer, J. R.: Insulin-binding sites in the rat brain: In vivo localization to the circumventricular organs by quantitative radioautography. *Endocrinology* 105; 666-673, 1979.